This is an amazing title

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Abstract

CCS Concepts

• **Computing methodologies** → **Natural language generation**;
• **Information systems** → **Question answering**.

ACM Reference Format:
Anonymous Author(s). 2026. This is an amazing title. In *Proceedings of The 35th International ACM Conference on Knowledge and Information Management (CiKM'26).* ACM, New York, NY, USA, 4 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Retrieval-Augmented Generation (RAG) based pipelines are one of the most common approaches for improving the grounding of Large Language Model (LLM) answers by conditioning generation on retrieved documents [4]. However, system-side grounding does not necessarily make an answer verifiable to an end user. An LLM-generated response may be faithful to retrieved evidence without exposing that evidence and, in contrast, a response may display citations that do not substantively support the claims to which they are attached. This distinction is relevant for information access systems, where users must often decide whether to trust or verify a concise generated answer under limited time and attention.

Prior work indicates that citations and explanations can substantially affect trust judgments, but not always in ways that align with evidential support. Ding et al. found that citations increased self-reported trust in LLM generated answers –even when the citations were random–, while users who checked citations reported lower trust [1]. Łajewska and Balog similarly showed that explanations for RAG responses can guide users toward higher-quality answers, but also that trust is mainly affected by clarity, actionability, topical context, and prior knowledge [3]. These findings indicate that the study of citations should be assessed beyond their evidential content and its role in enhancing perceived answer credibility, to also consider how different citation formats mediate the trade-off between cognitive effort and users' ability to assess answer trustworthiness.

This work focuses on *visual evidence attribution*: the user-oriented design of how retrieved evidence is connected to generated claims. The aim is to demonstrate how different evidence formats –such as inline citations, margin-linked evidence, end-of-answer source lists, document titles, links, snippets or highlights– shape the way users trust a generated answer. The core idea is not only to understand how credibility is impacted by presentation, but also to measure how this variations affect user satisfaction: what is the appropriate compromise between amount of evidence and users' cognitive load in order to fulfill a Question Answering task with evidence? While short cues may help improve readability by being straight to the point, lengthier snippets might increase perception of claim checking by providing more useful context –at the expense of escalate the effort needed to visually extract the important facts–.

We therefore aim to evaluate user interaction regulating two parameters: correctness (both answer and supporting evidence) and supporting evidence representation. A correct answer may be paired with incorrect, partial, or fully suported evidence, whereas an incorrect answer may be supported by sources that appear authoritative. Additionally, we modulate different formats and presentations, as explained previously in order to calibrate cognitive load. By treating credibility, cognitive load and satisfaction as measurable outcomes, the objective is to measure appropriate trust: measure whether confidence increases when claims are well supported, study the presence of overtrust when evidence is weak, but plausible; and checking when verification demands are cognitively burdensome, thus making users disengage from the evidence provided.

This user study investigates how alternative evidence formats affect perceived credibility, user satisfaction, and users' ability to discriminate supported from unsupported claims. The intended contribution an empirical account of how provenance design mediates the relationship between correctness, cognitive load, and calibrated trust in grounded LLM answers.

Guided by the distinction between answer correctness, evidential support, and user-facing presentation, we address the following research questions:

RQ1 How do different visual evidence attribution formats affect users' perceived credibility, satisfaction, and verification behavior when assessing RAG-generated answers?
RQ2 How do answer correctness and evidence support interact with presentation format to shape calibrated trust, including overtrust in weakly supported answers and undertrust in well-supported answers?
RQ3 What trade-offs emerge between evidence richness, evidence granularity, and cognitive load when users evaluate whether generated claims are supported by the displayed sources?

2 Related Work

2.1 Grounding, attribution, and verifiability

Technical work on RAG and grounded generation has primarily examined whether generated content is supported by retrieved evidence. This includes work on factual consistency, claim-level support, and attribution quality [6]. Such work is necessary for building reliable systems, since user-facing evidence is only meaningful when it genuinely supports the generated answer. However, this literature often treats evidence as an evaluation object rather

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CiKM'26, Rome, Italy
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ACM ISBN 978-1-4503-XXXX-X/2026/07
https://doi.org/XXXXXXX.XXXXXXX

than as an interaction object. From a user-study perspective, support must be both semantically valid and practically actionable: evidence that remains hidden, poorly located, or costly to interpret may provide limited value even when it is available to the system.

This separation motivates a distinction among source faithfulness, factual correctness, relevance, and citation quality. A faithful answer may omit explicit attribution, and a cited answer may still be unfaithful if the cited source is irrelevant, partial, or contradicted by other evidence. We use this distinction to motivate claim-level incentives in which answer correctness and support correctness can be manipulated independently from the visual format used to present evidence.

2.2 Citations, source presentation, and trust

Citations can increase trust in generated answers –even when they do not improve evidential quality–. Ding et al. manipulated citation presence, number, and relevance, showing that citations increased perceived trust even when they were random, and that checking citations was associated with lower trust [1]. This result highlights a risk for generative information access systems: citation visibility may operate as a generic credibility cue rather than as a reliable guide to support.

Recent work on source transparency further shows that citation layout can shape verification behavior. He and Liu compared conversational AI interfaces that expose sources through collapsible lists, hover cards, footer cards, and aligned sidebars, finding that source presentation changes exploration strategies and evidence integration [2]. This work demonstrates the importance of source-interface design, while leaving open the specific question we address: how evidence presentation affects credibility and satisfaction in claim-level RAG answers where answer correctness and support correctness can be independently controlled.

Our work extends this line of research by isolating the format through which evidence is exposed. A compact endnote, an inline claim-level citation, and a margin snippet may refer to the same source while imposing different verification costs and different credibility cues. We therefore evaluate whether visual evidence design affects both subjective credibility and behavioral checking, especially when evidence is partially supportive, irrelevant, or contradictory.

2.3 RAG explanations and user-facing evidence

Łajewska and Balog studied how source attribution, factual grounding, and information coverage affect trust judgments for RAG responses [3]. Their findings are directly relevant because they show that explanations can move users toward higher-quality responses, while also demonstrating that trust depends on response clarity, actionability, domain context, and prior knowledge. We build on this insight by shifting the unit of analysis from whether the explanation is present to evidence presentation format.

Other RAG-centered user studies also point to a trade-off between support for verification and cognitive effort. Reinhard et al. show that explanations can improve users' ability to identify factual confabulations in RAG-based LLM systems, but that some explanation types also increase time and cognitive effort [7]. This supports treating workload as a core outcome rather than as a secondary usability measure.

Rather than comparing explanation semantics alone, we aim to hold the underlying evidence constant and examine how its presentation affects trust, satisfaction and workload. This enables a direct test of design trade-offs: whether richer snippets improve correctness judgments at the cost of cognitive load; whether progressive disclosure preserves satisfaction while supporting verification; and whether claim-level attribution helps users identify unsupported claims more effectively than answer-level citation lists.

2.4 Cognitive burden and evidence granularity

Cognitive load can be understood as verification cost rather than merely as interface length. Work on human overreliance argues that users strategically decide whether engaging with an explanation is worth the effort under different conditions. Specifically, explanations are most useful when they reduce the cost of checking an AI output rather than simply adding more information [8]. This framing is directly applicable to evidence attribution: inline quotes, hover snippets, sidebars, and end-of-answer citations are distinct verification-cost structures, not just different visual styles.

Evidence granularity introduces an additional tension. PaperTrail decomposes LLM answers and sources into claim-evidence units for scholarly Q&A, lowering users' trust relative to a citation baseline but not necessarily changing reliance behavior because detailed checking remained cognitively burdensome [5]. This finding suggests that claim-level evidence may increase caution, but only improves interaction if the representation remains lightweight enough to support actual verification. This aligns with this works interest in finding which extractive evidence representations best calibrate credibility relative to correctness while maintaining acceptable cognitive load.

3 Methodology

4 Experimental Setup

5 Results

6 Conclusions

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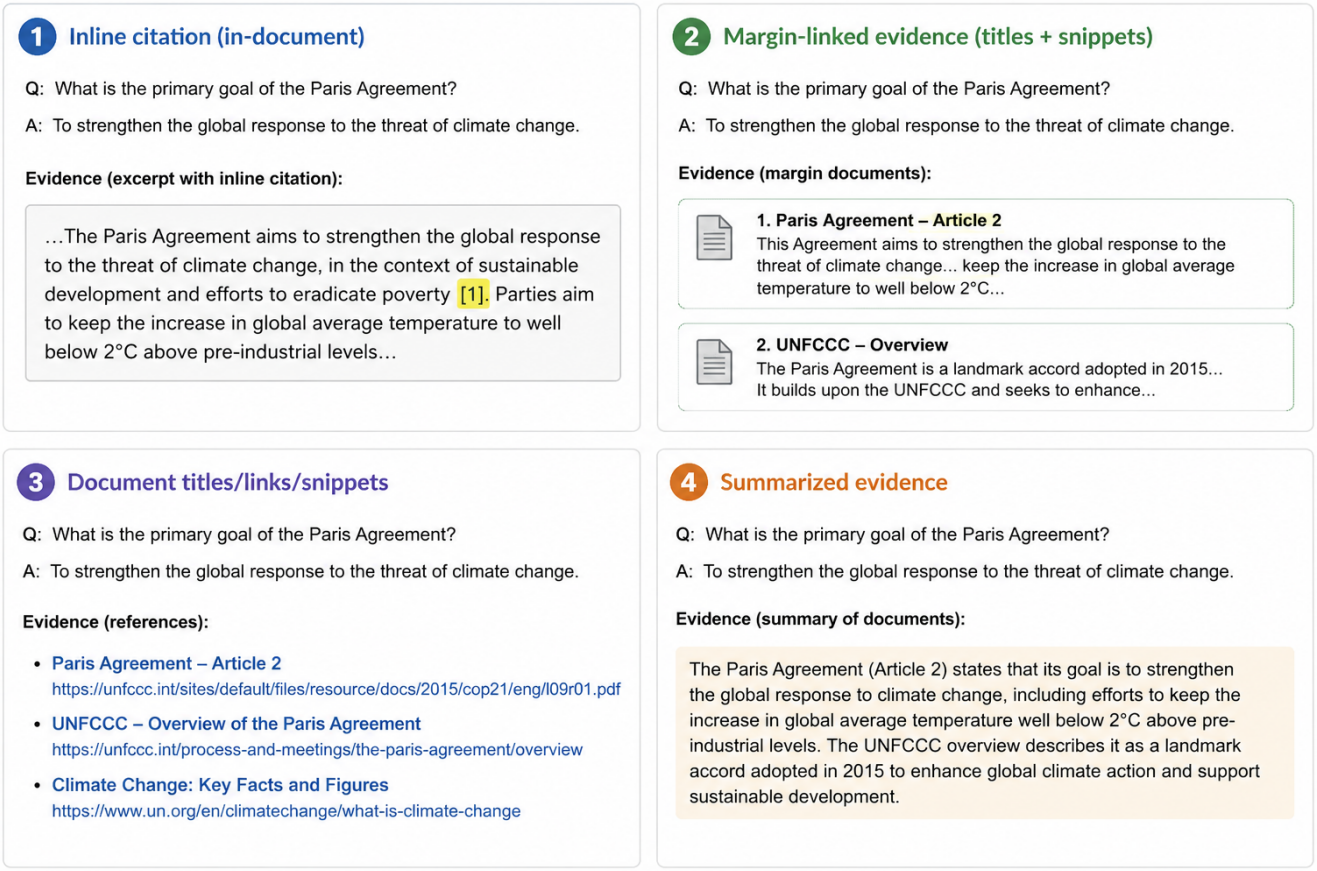


Figure 1: Comparison of evidence presentation formats in a question-answering system.

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Table 1: Variables considered in the online experiment.

Role	Variable	Operationalization	Levels or measurement
Independent	Answer correctness	Whether the generated answer correctly resolves the information need.	Correct; incorrect
Independent	Evidence support	Degree to which the displayed evidence substantively supports the claims in the generated answer.	Fully supporting; partially supporting; non-supporting or contradictory
Independent	Evidence placement	Location and visual connection between claims and their corresponding evidence.	Inline citations; margin-linked evidence; end-of-answer source list
Independent	Evidence detail	Amount and type of source information exposed to the user.	Citation marker only; title and link; source snippet; highlighted supporting passage
Independent	Visual claim–evidence linkage	Strength of the visual mapping between an answer claim and the evidence intended to support it.	No explicit linkage; citation-based linkage; direct or highlighted linkage
Dependent	Perceived credibility	User's judgment of how credible or trustworthy the generated answer appears.	Likert-scale rating
Dependent	Trust in the answer	Confidence that the answer is correct and can be relied upon.	Likert-scale rating or confidence probability
Dependent	User satisfaction	Overall satisfaction with the answer and the way supporting evidence is presented.	Likert-scale rating
Dependent	Perceived cognitive load	Subjective effort required to understand the answer and assess its supporting evidence.	Multi-item cognitive-load scale or Likert-scale rating
Dependent	Support discrimination	Ability to distinguish claims that are supported by the displayed evidence from claims that are unsupported.	Classification accuracy, sensitivity, specificity, or d'
Dependent	Verification accuracy	Correctness of the user's final judgment about whether the answer should be trusted.	Correct/incorrect judgment; proportion correct
Dependent	Verification effort	Behavioral effort invested in examining the available evidence.	Number of citations opened, snippets inspected, clicks, or dwell time
Dependent	Task completion time	Time required to inspect the answer and provide a trust or correctness judgment.	Time in seconds
Dependent	Evidence engagement	Extent to which the participant interacts with or attends to the supporting evidence.	Citation-opening rate, scrolling, highlighting, or evidence-viewing time
Derived outcome	Trust calibration	Alignment between the participant's reported trust and the actual correctness and evidential support of the answer.	Calibration error, Brier score, or trust-rating difference across conditions
Derived outcome	Overtrust	High reported trust in an incorrect answer or in an answer accompanied by weak or non-supporting evidence.	Trust above a predefined threshold in unsupported or incorrect conditions
Derived outcome	Undertrust	Low reported trust in a correct answer accompanied by fully supporting evidence.	Trust below a predefined threshold in fully supported conditions
Covariate	Prior topic knowledge	Participant's self-reported or objectively assessed familiarity with the question topic.	Self-report scale or knowledge-test score
Covariate	Citation literacy	Familiarity with interpreting citations and assessing whether sources support claims.	Self-report or citation-evaluation test
Covariate	LLM and RAG familiarity	Previous experience with LLM-based search, question answering, and citation-enabled assistants.	Frequency or experience scale
Covariate	General trust propensity	Participant's baseline tendency to trust automated systems or online information.	Validated trust-propensity scale
Control	Source authority	Apparent quality and reputation of the documents used as evidence.	Held constant or balanced across conditions
Control	Answer characteristics	Surface properties of generated responses that may affect trust independently of evidence.	Control answer length, tone, fluency, and number of claims
Control	Evidence quantity	Number of citations or evidence items presented with each answer.	Held constant unless explicitly manipulated
~~Control~~	~~Question characteristics~~	~~Properties of the information need that may influence difficulty and trust.~~	~~Balance topic, complexity, ambiguity, and expected prior knowledge~~
Control	Interface and exposure	Presentation conditions unrelated to the experimental manipulation.	Fixed layout, device requirements, time limit, and interaction instructions